

INDIAN INSTITUTE OF SCIENCE EDUCATION AND RESEARCH MOHALI

Mid-Term II 2025-2026

IDC 214: QUANTITATIVE ESTIMATES IN SOFT MATTER

DURATION 1 HR

FULL MARKS 15

NAME: SIDDHANT PANDA

SERIAL NUMBER: MS24129

DONOT write anything on the question paper.

1. For a fluorescence microscope using 500 nm excitation light and an objective with a numerical aperture (NA) of 1.6, calculate Abbe's lateral resolution limit. (2)
2. From your assignment, name any four (material) properties of the web that Spider-man needed. Make a Table as following and fill the respective boxes as marked.
(0.5 + 1 + 0.5) x 4 = 8

Name of the Material Property	Justification: Why is this important?	Any quantitative estimate

3. See the following Graph of E_{FRET} vs d (distance between dye pair). 2 + 2 + 1 = 5
 - (a). Define R_0 and estimate the value for this particular dye pair.
 - (b). Find out the distance between the dye-pair for E_{FRET} of 0.3.
 - (c). Justify how accurately you can determine the $d = 10$ nm of this dye-pair and why?

